Series : **2KNLM**

Set – **4**

प्रश्न-पत्र कोड
Q.P. Code **53**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।
Candidates must write the Q.P. Code on the title page of the answer-book.

कंप्यूटर एप्लीकेशन []

COMPUTER APPLICATIONS

निर्धारित समय : ***2*** *घण्टे* *अधिकतम अंक :* ***50***

Time allowed : ***2*** *hours* *Maximum Marks :* ***50***

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **19** हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **26** प्रश्न हैं।
- ***कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।***
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains **19** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **26** questions.
- **Please write down the Serial Number of the question in the answer-book at the given place before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.

सामान्य निर्देश :

कृपया निर्देशों को ध्यानपूर्वक पढ़िए :

(i) *इस प्रश्न-पत्र में **5** खण्ड हैं : खण्ड **क, ख, ग, घ, ङ** ।*

(ii) ***सभी** प्रश्न **अनिवार्य** हैं । यद्यपि कुछ प्रश्नों में आंतरिक विकल्प दिए गए हैं ।*

(iii) ***खण्ड क** में प्रत्येक **1** अंक के **12** प्रश्न हैं ।*

(iv) ***खण्ड ख** में प्रत्येक **2** अंक के **7** अति लघु-उत्तरीय (VSA) प्रकार के प्रश्न हैं ।*

(v) ***खण्ड ग** में प्रत्येक **3** अंक के **4** लघु-उत्तरीय (SA) प्रकार के प्रश्न हैं ।*

(vi) ***खण्ड घ** में प्रत्येक **4** अंक का **1** दीर्घ-उत्तरीय (LA) प्रकार का प्रश्न है ।*

(vii) ***खण्ड ङ** में प्रत्येक **4** अंक के स्रोत-आधारित/प्रकरण-आधारित/परिच्छेद-आधारित **2** प्रश्न हैं ।*

खण्ड क

(सभी प्रश्न अनिवार्य हैं) *12×1=12*

1. इंटरनेट क्या है ? *1*

(A) एक प्रकार का कंप्यूटर सॉफ़्टवेयर

(B) आपस में जुड़े कंप्यूटर नेटवर्कों का एक वैश्विक नेटवर्क

(C) किसी स्कूल में एक लोकल एरिया नेटवर्क (Local area network)

(D) एक प्रकार का कंप्यूटर हार्डवेयर

2. TCP/IP का प्राथमिक उद्देश्य क्या है ? *1*

(A) वेब पेजों का प्रबंधन (manage) करना

(B) सर्वर पर ईमेल स्टोर करना

(C) यह स्पष्ट करना कि इंटरनेट पर डेटा कैसे ट्रांसमिट (transmit) किया जाता है

(D) एन्क्रिप्शन (encryption) का उपयोग करके वेबसाइटों को सुरक्षित करना

3. कौन-सा हेडिंग टैग एचटीएमएल (HTML) में सबसे छोटे हेडिंग आकार को दर्शाता है ? *1*

(A) `<h1>` (B) `<h3>`

(C) `<h5>` (D) `<h6>`

4. किसी ईमेल एड्रेस में, उपयोगकर्ता का नाम (username) और डोमेन नाम (domain name) _____ प्रतीक (symbol) द्वारा अलग किए जाते हैं । *1*

(A) $ (B) @

(C) # (D) %

General Instructions :

Please read the instructions carefully.

(i) *This question paper has **5** Sections : Sections **A, B, C, D, E**.*

(ii) ***All** questions are **compulsory**. However, internal choices have been provided in some of the questions.*

(iii) ***Section A** has **12** questions carrying **1** mark each.*

(iv) ***Section B** has **7** Very Short Answer (VSA) type questions carrying **2** marks each.*

(v) ***Section C** has **4** Short Answer (SA) type questions carrying **3** marks each.*

(vi) ***Section D** has **1** Long Answer (LA) type question carrying **4** marks.*

(vii) ***Section E** has **2** Source-based / Case-based / Passage-based questions carrying **4** marks each.*

SECTION A
(All questions are compulsory) *12×1=12*

1. What is Internet ? *1*

(A) A type of computer software

(B) A global network of interconnected computer networks

(C) A local area network in a school

(D) A type of computer hardware

2. What is the primary purpose of the TCP/IP ? *1*

(A) To manage web pages

(B) To store emails on the server

(C) To define how data is transmitted over the Internet

(D) To secure websites using encryption

3. Which heading tag represents the smallest heading size in HTML ? *1*

(A) `<h1>` (B) `<h3>`

(C) `<h5>` (D) `<h6>`

4. In any email address, the username and the domain name are separated by _____ symbol. *1*

(A) $ (B) @

(C) # (D) %

5. निम्नलिखित में से कौन उन नियमों को दर्शाता है जो कंप्यूटरों को नेटवर्क पर आपस में संवाद करने में सक्षम बनाते हैं ? *1*

(A) ब्राउज़र
(B) ई-बैंकिंग
(C) प्रोटोकॉल
(D) टेक्स्ट एडिटर

6. POP3 प्रोटोकॉल में, संख्या "3" क्या दर्शाती है ? *1*

(A) यह शामिल तीन कंप्यूटरों को दर्शाता है
(B) यह वर्ष 2003 को दर्शाता है
(C) यह ईमेल डिलीवरी में उपयोग किए जाने वाले तीन मुख्य सर्वरों को दर्शाता है
(D) यह पोस्ट ऑफिस प्रोटोकॉल के तीसरे वर्ज़न (version) को दर्शाता है

7. निम्नलिखित में से कौन-सा एचटीएमएल (HTML) में List का मान्य प्रकार ***नहीं*** है ? *1*

(A) डेस्क्रिप्शन सूची (Description list)
(B) व्यवस्थित सूची (Ordered list)
(C) डिटेल सूची (Detail list)
(D) अव्यवस्थित सूची (Unordered list)

8. अजय, जो एक कंप्यूटर इंजीनियर है, सामान्यत: अपने लैपटॉप का उपयोग करके ग्राहकों के कंप्यूटरों को रीमोट लॉगिन से एक्सेस (access) करता है ताकि उनके कंप्यूटर की समस्याओं का पता लगा सके। इस उद्देश्य के लिए कौन-से प्रोटोकॉल का उपयोग हो रहा है ? *1*

(A) FTP
(B) SMS
(C) SMTP
(D) TELNET

9. वर्ल्ड वाइड वेब क्या है ? *1*

(A) एक नेटवर्क हार्डवेयर
(B) डेटा प्रबंधन सॉफ्टवेयर
(C) आपस में जुड़े वेब पेजों का सिस्टम
(D) वेब सर्वर

10. `<ol>` टैग (tag) में **`start`** ऐट्रिब्यूट (attribute) का क्या उद्देश्य है ? *1*

(A) बुलेट (bullet) के प्रकार को बदलना
(B) सूची (list) में आइटमों (items) की कुल संख्या निर्दिष्ट (specify) करना
(C) किसी अन्य सूची (list) के अंदर एक सूची (list) बनाना
(D) सूची (list) की प्रारंभिक संख्या सेट (set) करना

5. Which of the following refers to the rules that enable computers to communicate over a network ? *1*

(A) Browser (B) e-Banking

(C) Protocol (D) Text Editor

6. In the POP3 protocol, what does the number "3" signify ? *1*

(A) It represents three computers involved

(B) It stands for the year 2003

(C) It refers to three main servers used in email delivery

(D) It indicates the third version of the Post Office Protocol

7. Which of the following is ***not*** a valid type of list in HTML ? *1*

(A) Description list (B) Ordered list

(C) Detail list (D) Unordered list

8. Ajay, a computer engineer, generally uses his laptop to remotely access customers' computers in order to diagnose problems in their computer. Which protocol is being used for this purpose ? *1*

(A) FTP (B) SMS

(C) SMTP (D) TELNET

9. What is World Wide Web ? *1*

(A) A network hardware

(B) Software to manage data

(C) System of interlinked webpages

(D) Web server

10. What is the purpose of the `start` attribute in the `<ol>` tag ? *1*

(A) To change the type of bullets

(B) To specify the total number of items in the list

(C) To create a list within another list

(D) To set the starting number of the list

*प्रश्न संख्या **11** और **12** अभिकथन और कारण प्रकार के प्रश्न हैं । प्रत्येक प्रश्न में दो कथन हैं, अर्थात् अभिकथन (A) और कारण (R) । अभिकथन और कारण पर विचार करते हुए सबसे उपयुक्त विकल्प का चयन कीजिए ।*

(A) अभिकथन (A) और कारण (R) दोनों सही हैं और कारण (R), अभिकथन (A) की सही व्याख्या करता है ।

(B) अभिकथन (A) और कारण (R) दोनों सही हैं, परंतु कारण (R), अभिकथन (A) की सही व्याख्या ***नहीं*** करता है ।

(C) अभिकथन (A) सही है, परंतु कारण (R) ग़लत है ।

(D) अभिकथन (A) ग़लत है, परंतु कारण (R) सही है ।

11. *अभिकथन (A) :* ब्लॉग (Blog), व्यक्तियों के लिए अपने विचारों, ज्ञान और अनुभवों को ऑनलाइन साझा करने का एक प्रभावी तरीका है ।

कारण (R) : केवल पेशेवर वेब डेवलपर्स (professional web developers) ही ब्लॉग बना और उसका रख-रखाव (maintain) कर सकते हैं । *1*

12. *अभिकथन (A) :* **<img>** टैग, **src** ऐट्रिब्यूट (attribute) निर्दिष्ट (specify) न होने पर भी इमेज प्रदर्शित करेगा ।

कारण (R) : **src** ऐट्रिब्यूट (attribute) इमेज फ़ाइल को path प्रदान करती है, जो ब्राउज़र के लिए इमेज का पता लगाने और प्रदर्शित करने के लिए आवश्यक है । *1*

खण्ड ख *7×2=14*

13. सॉफ़्टवेयर लाइसेंस क्या है ? इसका उद्देश्य भी बताइए । *2*

14. (क) सर्च इंजन (search engine) क्या है ? किन्हीं दो लोकप्रिय सर्च इंजनों (search engines) के नाम बताइए । *2*

अथवा

(ख) SMTP का विस्तृत रूप लिखिए । SMTP, POP3 से किस प्रकार भिन्न है ? *2*

Questions No. ***11*** *and* ***12*** *are Assertion and Reason type questions. Each question consists of two statements, namely, Assertion (A) and Reason (R). Select the most suitable option considering the Assertion and Reason.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

11. *Assertion (A) :* Blogs are an effective way for individuals to share their thoughts, knowledge, and experiences online.

Reason (R) : Only professional web developers can create and maintain blogs. *1*

12. *Assertion (A) :* The **<img>** tag will display an image even if the **src** attribute is not specified.

Reason (R) : The **src** attribute provides the path to the image file, which is essential for the browser to locate and display the image. *1*

SECTION B

7×2=14

13. What is a software license ? Also mention its purpose. *2*

14. (a) What is a search engine ? Name any two popular search engines. *2*

OR

(b) Write the expanded form of SMTP. How does SMTP differ from POP3 ? *2*

15. एक उपयुक्त उदाहरण की सहायता से एचटीएमएल (HTML) में टैग (tag) और ऐट्रिब्यूट (attribute) के बीच अंतर स्पष्ट कीजिए। *2*

16. (क) विवरण सूची (description list) में **<DT>** और **<DD>** टैग्स (tags) का क्या उद्देश्य है ? *2*

अथवा

(ख) निम्नलिखित टैग्स (tags) का उद्देश्य स्पष्ट कीजिए : *2*

(i) **<P>**

(ii) **<HR>**

17. CSS क्या है और वेब डिजाइनिंग में इसका उपयोग क्यों किया जाता है ? *2*

18. (क) सोनल को एक अधूरा एचटीएमएल (HTML) कोड दिया गया है। दिए गए निर्देशों के अनुसार खाली स्थानों को भरकर कोड पूरा करने में उसकी सहायता कीजिए : *2*

```
<html>
<head>
<style>
p{
__________: Green;          /* Line 1 */
__________: 12px;           /* Line 2 */
__________: center;         /* Line 3 */
color:yellow;
}
__________ {                /* Line 4 */
width:50%
}
</style>
</head>
<body>
<p> Saving the environment is crucial for a healthier
Earth </p>
<hr>
</body>
</html>
```

15. Differentiate between a tag and an attribute in HTML with the help of a suitable example. *2*

16. (a) What is the purpose of **<DT>** and **<DD>** tags in a description list ? *2*

OR

(b) Explain the purpose of the following tags : *2*

(i) **<P>**

(ii) **<HR>**

17. What is CSS and why is it used in web designing ? *2*

18. (a) Sonal has been given an incomplete HTML code. Help her to complete the code by filling in the blanks as per the mentioned specifications : *2*

```
<html>
<head>
<style>
p{
__________: Green;          /* Line 1 */
__________: 12px;           /* Line 2 */
__________: center;         /* Line 3 */
color:yellow;
}
__________ {                /* Line 4 */
width:50%
}
</style>
</head>
<body>
<p> Saving the environment is crucial for a healthier
Earth </p>
<hr>
</body>
</html>
```

(i) Line 1 : पैराग्राफ का बैकग्राउंड रंग हरा (Green) होना चाहिए।

(ii) Line 2 : पैराग्राफ के टेक्स्ट (text) का फ़ॉन्ट आकार (size) 12 होना चाहिए।

(iii) Line 3 : पैराग्राफ के टेक्स्ट का संरेखण (alignment) center होना चाहिए।

(iv) Line 4 : क्षैतिज रेखा (horizontal ruler) की width 50% होनी चाहिए।

अथवा

(ख) नीचे दिए गए अधूरे एचटीएमएल (HTML) कोड का उल्लेख कीजिए और दिए गए निर्देशों के आधार पर इसे पूरा कीजिए। *2*

```
<html>
<head> __________</head>                <!-- Line 1 -->
<body>
<h1>My Favourite Fruits<h1>
<ol __________="I" __________="3">      <!-- Line 2 -->
__________                              <!-- Line 3 -->
<li>Banana</li>
</ol>
</body>
</html>
```

निम्नलिखित विनिर्देशों के अनुसार रिक्त स्थानों की पूर्ति कर इस कोड को पूरा कीजिए :

(i) Line 1 : वेब पेज का शीर्षक (title) **'About Me'** सेट कीजिए।

(ii) Line 2 : व्यवस्थित सूची (ordered list) में गिनती के लिए बड़े अक्षरों (uppercase) में रोमन अंकों (Roman numerals) का प्रयोग होना चाहिए।

(iii) Line 3 : सूची (list) की शुरुआत रोमन अंक III से होनी चाहिए।

(iv) Line 4 : सूची (list) का पहला आइटम (item) **'Apple'** होना चाहिए।

19. ई-बैंकिंग (e-Banking) क्या है ? ई-बैंकिंग (e-Banking) और ई-शॉपिंग (e-Shopping) में एक अंतर बताइए। *2*

(i) Line 1 : The background color of the paragraph should be Green.

(ii) Line 2 : The font size of the paragraph text should be 12.

(iii) Line 3 : The alignment of the paragraph text should be center.

(iv) Line 4 : The width of the horizontal ruler should be 50%.

OR

(b) Refer to the incomplete HTML code given below and complete the code based on the specifications provided. *2*

```
<html>
<head> __________</head>              <!-- Line 1 -->
<body>
<h1>My Favourite Fruits<h1>
<ol __________="I" __________="3">    <!-- Line 2 -->
__________                            <!-- Line 3 -->
<li>Banana</li>
</ol>
</body>
</html>
```

Complete this code by filling in the blanks as per the specifications that follow :

(i) Line 1 : Set the web page title to **'About Me'**.

(ii) Line 2 : The ordered list should use uppercase Roman numerals for numbering.

(iii) Line 3 : The list should begin with Roman numeral III.

(iv) Line 4 : The first list item should be **'Apple'**.

19. What is e-Banking ? Mention one difference between e-Banking and e-Shopping. *2*

खण्ड ग

4×3=12

20. निम्नलिखित एचटीएमएल कथन (HTML Statements) सही तरह से नहीं लिखे गए हैं । सही कथनों को रेखांकित (underlined) सुधारों के साथ पुन: लिखिए । प्रत्येक कथन का वांछित उद्देश्य उसके नीचे दिया गया है । *3*

(i) **Click here**
<!– – लिंक **https://ncert.nic.in** पर नेविगेट (navigate) होना चाहिए । – –>

(ii) **<ul type="O">**
<!– – सूची (List) में hollow circle को बुलेट के रूप में प्रयोग किया जाना चाहिए । – –>

(iii) **My Webpage**
<!– – फ़ॉन्ट Arial होना चाहिए तथा फ़ॉन्ट का आकार (size) 5 होना चाहिए । – –>

21. निम्नलिखित प्रश्नों के उत्तर दीजिए : *3*

(i) SMS का विस्तृत रूप लिखिए ।

(ii) FTP और SFTP का विस्तृत रूप लिखिए । फ़ाइलों को ट्रांसफर (transfer) करने के लिए SFTP, FTP से किस प्रकार भिन्न है ?

22. (क) रवि ने अपने मित्र के ग्रेजुएशन (Graduation) का जश्न मनाने (celebrate करने) के लिए एक आकर्षक वेब पेज तैयार किया है, जिसमें रंगीन चित्र और फेस्टिव टेक्स्ट (festive text) हैं । वेब पेज में, नीचे दिए गए निर्देशों के अनुसार, एक Celebratory Video जोड़ने के लिए उपयुक्त एचटीएमएल टैग्स (HTML tags) लिखने में उसकी मदद कीजिए : *3*

(i) वीडियो फ़ाइल का नाम **celebration.mp4** है ।

(ii) वीडियो के कंट्रोल्स (Play, Pause, Volume आदि) दिखाई देने चाहिए ।

(iii) वेब पेज लोड होते ही वीडियो स्वत: प्ले (play) होना चाहिए ।

अथवा

(ख) (i) HTML में लिंकिंग (linking) का महत्त्व समझाइए ।

(ii) निम्नलिखित आवश्यकताओं को पूरा करने के लिए उपयुक्त HTML टैग (tag) और सही ऐट्रिब्यूट्स (attributes) लिखिए :

- जो लिंक (link) अभी तक विज़िट (visit) नहीं किए गए हैं, वे नीले (Blue) रंग में दिखाई दें ।
- जो लिंक (link) पहले ही विज़िट (visit) किए जा चुके हैं, वे बैंगनी (Purple) रंग में दिखाई दें । *3*

SECTION C

4×3=12

20. The following HTML statements are not written properly. Rewrite the correct statements with underlined corrections. The desired purpose of each statement is mentioned under it. *3*

(i) `<a link="https://ncert.nic.in/">Click here</a>`

<!- - The link should navigate to `https://ncert.nic.in` - ->

(ii) `<ul type="O">`

<!- - The list should use hollow circle as bullet. - ->

(iii) `<font color=Arial size=5>My Webpage</font>`

<!- - The font should be Arial, and the size of the font should be 5. - ->

21. Answer the following questions : *3*

(i) Write the expanded form of SMS.

(ii) Expand FTP and SFTP. How does SFTP differ from FTP for transferring files ?

22. (a) Ravi has designed a vibrant web page filled with colourful images and festive text to celebrate his friend's graduation. Help him by writing suitable HTML tag(s) to add a celebratory video to his web page as per the specification given below : *3*

(i) Name of the video file is `celebration.mp4`.

(ii) The video controls (Play, Pause, Volume, etc.) should be visible.

(iii) The video should automatically start playing when the web page loads.

OR

(b) (i) Explain the significance of linking in HTML.

(ii) Write the appropriate HTML tag with the correct attributes to achieve the following requirements :

- The links that have not been visited should appear in blue.
- The links that have already been visited should appear in purple. *3*

23. (क) उपयुक्त एचटीएमएल कथन (HTML statements) लिखिए जिससे नीचे दिया गया टेक्स्ट वेब पेज पर दिखाई दे : *3*

(i) H_2O

(ii) $E=mc^2$

(ख) हाइपरलिंक (Hyperlink) बनाने हेतु एचटीएमएल कोड (HTML code) लिखिए ताकि जब उपयोगकर्ता **'HELP'** टेक्स्ट (text) पर क्लिक करें, तो **"help@cbse.gov.in"** पर ईमेल भेजा जा सके।

खण्ड घ

24. (क) नीचे दिखाए गए फॉर्म के साथ वेब पेज डिज़ाइन करने के लिए HTML कोड लिखिए : *4*

Flight Reservation Form

Enter Name : [ ]

Seat Type : ◉ Economy ○ Business

Select Category : Child ∨

- Child
- Adult
- Senior Citizen

कोड लिखते समय निम्नलिखित विनिर्देशों (specifications) पर ध्यान दीजिए :

- **Specification 1 :** टेक्स्ट **"Flight Reservation Form"** को heading level 3 में दिखाया जाना चाहिए।
- **Specification 2 :** यात्री (Passenger) का नाम दर्ज करने के लिए एक Text Box होना चाहिए।
- **Specification 3 :** सीट टाइप (Seat Type) (Economy या Business) चुनने के लिए दो Radio Buttons होने चाहिए, जैसा कि ऊपर दिखाया गया है। इनमें से Economy डिफ़ॉल्ट रूप से सेलेक्टेड (selected) होना चाहिए।
- **Specification 4 :** यात्री की कैटेगरी (Category) चुनने के लिए एक Combo Box होना चाहिए, जिसमें Child, Adult, और Senior Citizen विकल्प (option) हों।

अथवा

23. (a) Write suitable HTML statements to display the following text on a webpage : *3*

(i) H_2O

(ii) $E=mc^2$

(b) Write HTML code to create a hyperlink for sending an email to **"help@cbse.gov.in"** when the user clicks on the text **'HELP'**.

SECTION D

24. (a) Write the HTML code to design the webpage with the form as shown below : *4*

Flight Reservation Form

Enter Name : [ ]

Seat Type : ◉ Economy ○ Business

Select Category : Child ∨

- Child
- Adult
- Senior Citizen

Please note the following specifications while writing the code :

- **Specification 1 :** The text **"Flight Reservation Form"** should be in heading level 3.
- **Specification 2 :** There should be a text box to accept name of the passenger.
- **Specification 3 :** Two radio buttons to choose the seat type (Economy or Business) as shown above. Economy should be the default selected radio button.
- **Specification 4 :** Passenger's category to be chosen from a combo box having options as Child, Adult, and Senior Citizen.

OR

(ख) दिए गए विनिर्देशों (specifications) को ध्यान में रखते हुए नीचे दिखाए अनुसार वेब पेज डिज़ाइन करने के लिए HTML कोड लिखिए : *4*

My Schedule

Name	Morning	Noon	Evening
Sahil	Running	Self Study	Gym
Abhay	Yoga	Tuition	
Neha	School		Dancing

- **Specification 1 :** वेब पेज का कोड सही HTML संरचना (structure) का पालन करे । **`"My Schedule"`** को पहले स्तर की heading का उपयोग करके दिखाया जाना चाहिए, और पेज का बैकग्राउंड रंग Light Yellow होना चाहिए।
- **Specification 2 :** कॉलम हेडर्स (column headers) के लिए टेबल हेडर टैग (table header tag) का उपयोग किया जाना चाहिए।
- **Specification 3 :** टेबल में कुल 4 पंक्तियाँ (rows) और 4 कॉलम (columns) होने चाहिए । पहली पंक्ति में टेबल हेडर (table header) प्रदर्शित होगा और बाकी पंक्तियाँ छात्रों (students) का डेटा दिखाने के लिए होंगी, जैसा कि ऊपर दिखाया गया है।
- **Specification 4 :** टेबल में दिए गए टेबल के अनुसार, जहाँ आवश्यक हो, वहाँ rowspan और colspan का उपयोग किया जाना चाहिए।

खण्ड ङ

केस-स्टडी *2×4=8*

25. डॉ. चित्रा एक विश्वविद्यालय में कंप्यूटर विज्ञान की प्रोफेसर हैं और पिछले कुछ महीनों से एक शोध पत्र पर काम कर रही थीं । हाल ही में उन्होंने अपना शोध पत्र एक प्रकाशन गृह को भेजा है । शोध पत्र के प्रकाशित होने के कुछ महीनों बाद उन्हें पता चला कि एक अन्य शोधकर्ता ने, एक अलग पत्रिका में, समान शोध पत्र प्रकाशित किया है, जिसमें उनके काम के कुछ हिस्सों का उपयोग बिना उन्हें श्रेय दिए किया गया है।

डॉ. चित्रा बहुत परेशान हैं और कानूनी कार्रवाई करना चाहती हैं।

उपर्युक्त जानकारी के आधार पर निम्नलिखित प्रश्नों के उत्तर दीजिए : *4*

(i) क्या इस दिए गए परिदृश्य में इंटरनेट शिष्टाचार (Netiquette) का पालन किया जा रहा है ? अपने उत्तर का औचित्य सिद्ध कीजिए।

(b) Write the HTML code to design the web page as shown below considering the given specifications : *4*

My Schedule

Name	Morning	Noon	Evening
Sahil	Running	Self Study	Gym
Abhay	Yoga	Tuition	
Neha	School		Dancing

- **Specification 1 :** The webpage code should follow the correct HTML structure. **`"My Schedule"`** should be displayed using the first level heading, and the background colour of the page should be light yellow.
- **Specification 2 :** Use the table header tag for the column headers.
- **Specification 3 :** The table should have exactly 4 rows and 4 columns, with the first row displaying the table header and the remaining rows for student data, as shown above.
- **Specification 4 :** The table should use rowspan and colspan, wherever required, as shown in the given table.

SECTION E
Case-Study

2×4=8

25. Dr. Chitra is a Computer Science Professor at a University and was working on a research paper since last few months. She recently submitted her paper to a publication house. A few months after publishing her paper, she found that another researcher has published a similar paper in a different journal, using some portions of her work without giving her credit.
Dr. Chitra is upset and wants to take legal action.

Based on the above information, answer the following questions : *4*

(i) Are netiquettes being followed in the given scenario ? Justify your answer.

(ii) किसी अन्य के कार्य को अपना बताकर उपयोग करने के कृत्य को _____ कहा जाता है।

(iii) डॉ. चित्रा की बिना अनुमति लिए उनके कार्य को प्रकाशित करके उस शोधकर्ता ने किस प्रकार के बौद्धिक संपदा कानून का उल्लंघन किया ?

(iv) शैक्षणिक अनुसंधान के संदर्भ में बौद्धिक संपदा कानून के उल्लंघन को रोकने का कोई एक उपाय सुझाइए।

26. इरफान एक ग्रामीण क्षेत्र में रहता है जहाँ इंटरनेट की सुविधा सीमित है। उसके बच्चे ऑनलाइन कक्षाओं में उपस्थित होने में कठिनाई का सामना करते हैं, क्योंकि उस क्षेत्र में इंटरनेट कनेक्टिविटी बहुत खराब है। दूसरी ओर, पास के शहर में लोगों को हाई-स्पीड इंटरनेट उपलब्ध है और वे डिजिटल सेवाओं का प्रभावी ढंग से उपयोग कर पाते हैं। इरफान को लगता है कि डिजिटल ढाँचे की कमी के कारण उसका परिवार पिछड़ रहा है।

उपर्युक्त जानकारी के आधार पर निम्नलिखित प्रश्नों के उत्तर दीजिए : *4*

(क) डिजिटल तकनीक तक असमान पहुँच को _____ कहा जाता है।

(ख) तकनीक तक असमान पहुँच का कोई एक कारण बताइए।

(ग) इस असमान पहुँच को कम करने के लिए सरकार द्वारा उठाए जा सकने वाले किसी एक कदम का उल्लेख कीजिए।

(घ) इंटरनेट की सीमित पहुँच, शिक्षा और अवसरों की दृष्टि से बच्चों को किस प्रकार प्रभावित करती है ?

(ii) The act of using someone's work as your own is known as _____.

(iii) Which type of intellectual property law was violated by the other researcher by publishing Dr. Chitra's work without consent ?

(iv) Suggest any one way to prevent intellectual property law violation in the context of academic research.

26. Irfan lives in a rural area with limited access to the Internet. His children struggle to attend online classes, as the area has poor Internet connectivity. On the other hand, in the nearby city, people have access to high speed Internet and are able to utilize services effectively. Irfan feels that his family is at a disadvantage due to the lack of access to digital infrastructure.

Based on the above information, answer the following questions : *4*

(a) Unequal access to digital technology is called _____.

(b) Mention any one reason for this unequal access to technology.

(c) Mention any one initiative the government can take to reduce this unequal access.

(d) How does limited access to the Internet affect children in terms of education and opportunities ?

Marking Scheme
Strictly Confidential
(For internal and restricted use only)
Secondary School Examination, 2026
SUBJECT NAME COMPUTER APPLICATIONS (Q.P. CODE 53)

General Instructions:		
1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.	
2	"Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its' leakage to the public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc. may invite action under various rules of the Board and IPC."	
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. However, while evaluating answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due Marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand the given answer and even if the reply is not from the Marking scheme but correct competency is enumerated by the candidate, due Marks should be awarded.	
4	The Marking Scheme carries only suggested value points for the answers. These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due Marks should be awarded accordingly.	
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the Marking of individual evaluators.	
6	Evaluators will Mark(√) wherever the answer is correct. For wrong answer CROSS 'X" be Marked. Evaluators will not put right (✓)while evaluating which gives an impression that the answer is correct and no Marks are awarded. This is the most common mistake which evaluators are committing.	
7	If a question has parts, please award Marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.	
8	If a question does not have any parts, Marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.	
9	If a student has attempted an extra question, the answer of the question deserving more Marks should be retained and the other answer scored out with a note "Extra Question".	
10	No Marks to be deducted for the cumulative effect of an error. It should be penalized only once.	

11	A full scale of Marks **50 Marks** as given in Question Paper) has to be used. Please do not hesitate to award full Marks if the answer deserves it.	
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).This is in view of the reduced syllabus and number of questions in question paper.	
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- ● Leaving the answer or part thereof unassessed in an answer book. ● Giving more Marks for an answer than assigned to it. ● Wrong totaling of Marks awarded on an answer. ● Wrong transfer of Marks from the inside pages of the answer book to the title page. ● Wrong question wise totaling on the title page. ● Wrong totaling of Marks of the two columns on the title page. ● Wrong grand total. ● Marks in words and figures not tallying/not same. ● Wrong transfer of Marks from the answer book to online award list. ● Answers Marked as correct, but Marks not awarded. (Ensure that the right tick Mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answers.) ● Half or a part of the answer Marked correct and the rest as wrong, but no Marks awarded.	
14	While evaluating the answer books, if the answer is found to be totally incorrect, it should be Marked as cross (X) and awarded zero (0)Marks.	
15	Any unassessed portion, non-carrying over of Marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.	
16	The Examiners should acquaint themselves with the guidelines given in the “Guidelines for Spot Evaluation” before starting the actual evaluation.	
17	Every Examiner shall also ensure that all the answers are evaluated, Marks carried over to the title page, correctly totaled and written in figures and words.	
18	The candidates are entitled to obtain a photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.	

MARKING SCHEME
COMPUTER APPLICATIONS

Max. Marks: 50

General Instructions :

Please read the instructions carefully.

*(i) This question paper has **5** Sections: Sections **A, B, C, D, E**.*

***(ii) All** questions are **compulsory**. However, internal choices have been provided in some of the Questions.*

*(iii) **Section A** has **12** questions carrying **1** mark each.*

*(iv) **Section B** has **7** Very Short Answer (VSA) type questions carrying **2** marks each.*

***(v) Section C** has **4** Short Answer (SA) type questions carrying **3** marks each.*

***(vi) Section D** has **1** Long Answer (LA) type question carrying **4** marks.*

*(vii) **Section E** has **2** Source-based/Case-based/Passage-based questions carrying **4** marks each.*

SECTION A **(All questions are compulsory)**					**12x1=12**
1.	What is Internet?				1
	(A)	A type of computer software	(B)	A global network of interconnected computer networks	
	(C)	A local area network in a school	(D)	A type of computer hardware	
Ans	(B) A global network of interconnected computer networks				
	(1 Mark for writing the correct option)				
2.	What is the primary purpose of the TCP/IP ?				1
	(A)	To manage web pages			
	(B)	To store emails on the server			
	(C)	To define how data is transmitted over the Internet			
	(D)	To secure websites using encryption			
Ans	(C) To define how data is transmitted over the Internet				
	(1 mark for writing the correct option)				
3.	Which heading tag represents the smallest heading size in HTML ?				1
	(A)	`<h1>`	(B)	`<h3>`	
	(C)	`<h5>`	(D)	`<h6>`	
Ans	(D) `<h6>`				
	(1 mark for writing the correct option)				

4.	In any email address, the username and the domain name are separated by _____ symbol.				1
	(A)	**$**	(B)	@	
	(C)	**#**	(D)	%	
Ans	(B) @				
	(1 mark for writing the correct option)				
5.	Which of the following refers to the rules that enable computers to communicate over a network ?				1
	(A)	Browser	(B)	e-Banking	
	(C)	Protocol	(D)	Text Editor	
Ans	(C) Protocol				
	(1 mark for writing the correct option)				
6.	In the POP3 protocol, what does the number "3" signify ?				1
	(A)	It represents three computers involved	(B)	It stands for the year 2003	
	(C)	It refers to three main servers used in email delivery	(D)	It indicates the third version of the Post Office Protocol	
Ans	(D) It indicates the third version of the Post Office Protocol				
	(1 mark for writing the correct option)				
7.	Which of the following is ***not*** a valid type of list in HTML ?				1
	(A)	Description list	(B)	Ordered list	
	(C)	Detail list	(D)	Unordered list	
Ans	(C) Detail list				
	(1 mark for writing the correct option)				
8.	Ajay, a computer engineer, generally uses his laptop to remotely access customers' computers in order to diagnose problems in their computer. Which protocol is being used for this purpose ?				1
	(A)	FTP	(B)	SMS	
	(C)	SMTP	(D)	TELNET	
Ans	(D) TELNET				
	(1 mark for writing the correct option)				

9.	What is World Wide Web ?				1
	(A)	A network hardware	(B)	Software to manage data	
	(C)	System of interlinked webpages	(D)	Web server	
Ans	(C) System of interlinked webpages				
	(1 mark for writing the correct option)				
10.	What is the purpose of the **`start`** attribute in the **`<ol>`** tag ?				1
	(A)	To change the type of bullets	(B)	To specify the total number of items in the list	
	(C)	To create a list within another list	(D)	To set the starting number of the list	
Ans	(D) To set the starting number of the list				
	(1 mark for writing the correct option)				

*Questions No. **11** and **12** are Assertion and Reason type questions. Each question consists of two statements, namely, Assertion (A) and Reason (R). Select the most suitable option considering the Assertion and Reason.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

11.	Assertion (A) : Blogs are an effective way for individuals to share their thoughts, knowledge, and experiences online. Reason (R) : Only professional web developers can create and maintain blogs.	1
Ans	(C) Assertion (A) is true, but Reason (R) is false.	
	(1 mark for writing the correct option)	
12.	Assertion (A):The **`<img>`** tag will display an image even if the **`src`** attribute is not specified. Reason (R) : The **`src`** attribute provides the path to the image file, which is essential for the browser to locate and display the image.	1
Ans	(D) Assertion (A) is false, but Reason (R) is true.	
	(1 mark for writing the correct option)	

SECTION B

7x2=14

13.	What is a software license ? Also mention its purpose.	2
Ans	A software license is a legal agreement that defines how a user can install, use, and distribute a software program **OR** Any other statement conveying similar meaning.	

Its purpose is to protect the rights of the software creator and to specify the terms under which the software can be legally used or shared.

(1 mark for writing correct definition of software license)
(1 mark for writing correct purpose)

14. (a) What is a search engine ? Name any two popular search engines. 2

Ans

A search engine is a online service/software/app that helps to find information on the internet by entering keywords or phrases.
OR
A search engine retrieves and displays a list of web pages that match the search query.

Examples (Any two) – Google, Bing, Yahoo, DuckDuckgo, AOL search, Ask.com , Yandex, Naver, Baidu, ChatGPT, Gemini, Perplexity

(1 mark for writing the correct definition of search engine)
(½ mark each for writing names of any two correct search engines)

OR

(b) Write the expanded form of SMTP. How does SMTP differ from POP3 ? 2

Ans

SMTP	**POP3**
Simple Mail Transfer Protocol	Post Office Protocol version 3
Is used to access emails on browser.	Is used to retrieve emails through email client software/apps
Does not save emails on user's device.	Downloads/Saves emails on user's device.

(1 mark for writing any one correct difference from the above mentioned/similar)

Note: 1 mark to be awarded if only fullform of SMTP is given

15. Differentiate between a tag and an attribute in HTML with the help of a suitable example. 2

Ans

Tag	**Attribute**
A tag is an HTML command that defines an element of a webpage.	An attribute provides additional information about a tag.
A tag creates and structures content like heading, paragraphs, images, hyperlink etc..	It modifies or gives extra details to the elements.
A tag is written inside angle brackets < >	It is written inside a tag.
Example: `<body bgcolor="red"> </body>` In this `<body>...</body>` is a tag. **OR**	Example: `<body bgcolor="red"> </body>` In this `bgcolor` is an attribute. **OR**

	<table><tr><td>**Hello** In this **...** is a tag. **OR** Any other equivalent example</td><td>**Hello** In this **face** is an attribute. **OR** Any other equivalent example</td></tr></table>	
	(1 mark for writing any one correct difference) ***(1 mark for writing any one correct example)*** ***OR*** ***(Full 2 marks for illustrating the difference using a correct example)***	
16.	(a) What is the purpose of **<DT>** and **<DD>** tags in a description list ?	2
Ans	**Purpose of <DT> Tag:** It is used to define the term or name in a description list. **OR** **<DT>** stands for Description Term. **Purpose of <DD> Tag:** It provides the description or explanation of the term defined by <DT>. **OR** **<DD>** stands for Description Definition.	
	(1 mark each for writing appropriate purpose/correct fullform)	
	OR (b) Explain the purpose of the following tags : (i) **<P>** (ii) **<HR>**	2
Ans	(i) **<P>** used to define a paragraph in HTML. **OR** **<P>** is a Paragraph tag **OR** **<P>** groups text into blocks and provides space before and after the paragraph to separate it from other content. (ii) **<HR>** is used to create horizontal line or rule across the page. **OR** **<HR>** is a Horizontal Rule tag. **OR** **<HR>** is used to separate contents and sections within the page.	
	(1 mark each for writing appropriate purpose/correct fullform)	
17.	What is CSS and why is it used in web designing ?	2
Ans	CSS stands for Cascading Style Sheets. Purpose: CSS is a style sheet language used to control the presentation of Web Page **OR** CSS is used to style and format the content of a webpage **OR** It is used for changing appearance and layout of content of a webpage **OR**	

	It is used for designing, control colours, fonts, spacing, alignment, positioning, and overall layout. **OR** Any other correct purpose	
	(2 marks for writing for any one appropriate use of CSS) ***OR*** ***(1 mark for writing only the correct fullform of CSS)***	

18.	(a) Sonal has been given an incomplete HTML code. Help her to complete the code by filling in the blanks as per the mentioned specifications :
	
	**<html>
	<head>
	<style>
	p{
	_____: Green; /* Line 1 */
	_____: 12px; /* Line 2 */
	_____: center; /* Line 3 */
	color:yellow;
	}
	_____ { /* Line 4 */
	width:50%
	}
	</style>
	</head>
	<body>
	 Saving the environment is crucial for a healthier Earth
	<hr>
	</body>
	</html>**
	
	(i) Line 1 : The background color of the paragraph should be Green.
	(ii) Line 2 : The font size of the paragraph text should be 12.
	(iii) Line 3 : The alignment of the paragraph text should be center.
	(iv) Line 4 : The width of the horizontal ruler should be 50%.
		2

Ans	(i) Line 1 : **background-color** (ii) Line 2 : **font-size** (iii) Line 3 : **text-align** (iv) Line 4 : **hr**	
	(½ mark each for writing correct missing code for the blanks)	
	OR	

	(b) Refer to the incomplete **HTML** code given below and complete the code based on the specifications provided.
	`<html>`
	`<head> _____</head>` `<!-- Line 1 -->`
	`<body>`
	`<h1>My Favourite Fruits<h1>`
	`<ol _____="I" _____="3">` `<!-- Line 2 -->`
	`_____` `<!-- Line 3 -->`
	`<li>Banana</li>`
	`</ol>`
	`</body>`
	`</html>`
	Complete this code by filling in the blanks as per the specifications that follow :
	(i) Line 1 : Set the web page title to **'About Me'**.
	(ii) Line 2 : The ordered list should use uppercase Roman numerals for numbering.
	(iii) Line 3 : The list should begin with Roman numeral **III**.
	(iv) Line 4 : The first list item should be **'Apple'**.
		2

Ans	(i) Line 1 : **<title> About Me </title>** (ii) Line 2 : **type** (iii) Line 3 : **start** (iv) Line 4 : ** - Apple ** **OR** (i) Line 1 : **<title> About Me </title>** (ii) Line 2 : **type start** (iii) Line 3 : ** - Apple **	
	(2 mark for writing the correct entry for any 3 missing part of the code) ***OR*** ***(1½ mark for writing the correct entry for any 2 missing part of the code)*** ***OR*** ***(1 mark for writing the correct entry for any 1 missing part of the code)***	
19.	What is e-Banking ? Mention one difference between e-Banking and e-Shopping.	2
Ans	e-banking stands for electronic banking. **OR** e-banking is online banking. **OR** e-banking enables customers to access and utilize banking services over the internet. **Difference:**	

e-Banking	**e-Shopping**
e-Banking is electronic banking	e-Shoping is electronic shopping
e-Banking means online banking service provided by banks through internet	e-Shopping means buying or selling goods or services through online websites or Apps
e-Banking allows financial transactions online	e-Shopping enables the purchase of goods or services online
provided by the banks	provided by online stores or e-commerce companies

	Example: transferring money using net banking **OR** onlinesbi.sbi **OR** sbi.bank.in **OR** icici.bank.in **OR** Any other e-Banking website/app	Example: ordering a mobile phone from an online shopping site **OR** bigbasket.com **OR** amazon.in **OR** esaras.in **OR** Any other e-Shopping website/app
	(2 mark for writing any one correct difference OR 1 mark for explaining any one) ***OR*** ***(1 mark for writing correct definition of e-banking)*** ***OR*** ***(½ mark each for writing correct example of e-banking / e-shopping)***	
	SECTION C *4x3=12*	

20.	The following **HTML** statements are not written properly. Rewrite the correct statements with underlined corrections. The desired purpose of each statement is mentioned under it.
	(i)
	`<a link="https://ncert.nic.in/">Click here</a>`
	`<!-- The link should navigate to https://ncert.nic.in -->`
	
	(ii)
	`<ul type="O">`
	`<!-- The list should use hollow circle as bullet. -->`
	
	(iii)
	`<font color=Arial size=5>My Webpage</font>`
	`<!-- The font should be Arial, and the size of the font should be 5. -->`
		3

Ans	(i) `<a href="https://ncert.nic.in/">Click here</a>`
	OR
	`<a href=https://ncert.nic.in/>Click here</a>`
	(ii) `<ul type="circle">`
	OR
	`<ul type=circle>`
	
	(iii) `<font face=Arial size=5> My Webpage </font>`
	OR
	`<font face="Arial" size="5"> My Webpage </font>`	

	(1 mark for writing each correction) ***OR*** ***(½ mark each to be awarded for writing only*** `href` ***or*** `circle` ***or*** `face`***)*** ***Note: underline to be ignored***	

21.	Answer the following questions: (i) Write the expanded form of SMS. (ii) Expand FTP and SFTP. How does SFTP differ from FTP for transferring files ?	3
Ans	(i) SMS: Short Message Service (ii) FTP : File Transfer Protocol SFTP : Secure File Transfer Protocol SFTP encrypts data before sending it over a network while FTP transfers data without encryption.	
	(1 mark for writing correction expansion of SMS) ***(1 mark each for writing correct expansion of FTP and SFTP)*** ***OR*** ***(1 mark for writing any correct difference OR ½ mark for explaining any one)***	
22.	(a) Ravi has designed a vibrant web page filled with colourful images and festive text to celebrate his friend's graduation. Help him by writing suitable **`HTML`** tag(s) to add a celebratory video to his web page as per the specification given below : (i) Name of the video file is **`celebration.mp4`**. (ii) The video controls (Play, Pause, Volume, etc.) should be visible. (iii) The video should automatically start playing when the web page loads.	3

Ans	`<video controls autoplay muted>`
	`<source src="celebration.mp4">`
	`</video>`
	OR
	`<video src="celebration.mp4" controls autoplay muted>`
	`</video>`
	OR
	`<embed src="celebration.mp4" autostart="true">`	

	(1 mark for correct use of `src` attribute) ***(1 mark for correctly displaying `controls` - marks to be awarded if `embed` tag is used without mention of `controls`)*** ***(1 mark for correct use of `autoplay/autostart` attribute)*** ***Note: Including the `muted` attribute is not a requirement for grading purpose.*** ***Important information:*** ***The following four points are just for information:*** ● *The autostart attribute for the <embed> tag is no longer supported in modern web browsers.* ● *Due to updated security and user experience policies, browsers now block videos with sound from playing automatically.* ● *To ensure a video starts on page load, the autoplay and muted attributes must both be used.*	
	OR	
	(b) (i) Explain the significance of linking in HTML. (ii) Write the appropriate HTML tag with the correct attributes to achieve the following requirements : • The links that have not been visited should appear in blue. • The links that have already been visited should appear in purple.	3
Ans	(i) Linking in HTML connects web pages and other web resources **OR**	

	Linking in HTML helps to navigate easily and makes websites interactive.
	
	(ii) `<body link="blue" vlink="purple">`
	`</body>`
	OR
	`<style>`
	`a:link {color: blue;}`
	`a:visited {color: purple;}`
	`</style>`
	OR
	`<a href="" style="link:blue, alink:purple">`
	OR
	Any other equivalent	

	(1 mark for writing correct significance of linking in HTML) ***(1 mark each for incorporating each specification)***	
23.	(a) Write suitable **HTML** statements to display the following text on a webpage : (i) H_2O (ii) $E=mc^2$	3
Ans	(i) `H<sub>2</sub>O` (ii) `E=mc<sup>2</sup>`	
	(1 mark for writing each correct statement) ***OR*** ***(½ mark each for mentioning*** `sub` ***or*** `sup`***)***	
	(b) Write HTML code to create a hyperlink for sending an email to `"help@cbse.gov.in"` when the user clicks on the text `'HELP'`.	
Ans	`<a href="mailto:help@cbse.gov.in">HELP</a>`	
	(1 mark for writing the correct code) ***OR*** ***(½ mark for only mentioning*** `a` ***tag)***	
	SECTION D	
24.	(a) Write the HTML code to design the webpage with the form as shown below : **Flight Reservation Form** Enter Name: [ ] Seat Type: (•) Economy ( ) Business Select Category: [Child ▽] Child Adult Senior Citizen Please note the following specifications while writing the code :	4

- **Specification 1 :** The text **"Flight Reservation Form"** should be in heading level 3.
- **Specification 2 :** There should be a text box to accept name of the passenger.
- **Specification 3 :** Two radio buttons to choose the seat type (Economy or Business) as shown above. Economy should be the default selected radio button.
- **Specification 4 :** Passenger's category to be chosen from a combo box having options as Child, Adult, and Senior Citizen.

Ans

```
<h3>Flight Reservation Form</h3>          <!--Specification 1 -->
<form>
 Enter Name :
 <input type="text">                      <!--Specification 2 -->
 Seat Type :                              <!--Specification 3 -->
 <input type="radio" name="SeatType" value="Economy" checked>
 Economy
 <input type="radio" name="SeatType" value="Business">Business

 <!-- Option 1 -->
 Select Category:
 <select id="Category" name="Category"> <!--Specification 4 -->
 <option value="Child">Child</option>
 <option value="Adult">Adult</option>
 <option value="Senior">Senior Citizen</option>
 </select>

</form>
```

```
 <!-- Option 2 -->
 Select Category:                        <!--Specification 4 -->
 <input type="checkbox" name="CatChild" value="Child">Child
 <input type="checkbox" name="CatAdult" value="Adult">Adult
 <input type="checkbox" name="CatSenior" value="Senior">Senior
Citizen
 </select>
```

Note: Any of the correct code out of option 1 or option 2 or any other equivalent code for Specification 4 should be considered as correct.

(1 mark for the heading using <h3> tag)
(1 mark for writing correct code for generating Textbox)
(1 mark for writing correct code for generating Radio buttons)
(1 mark for writing correct code for generating Combo Box/ List box)

Note:
The id, name, and value attributes are generally optional and will not impact grade, with one exception: per Specification 3, the name attribute is required for radio buttons to ensure they function correctly as a group.

4

OR

(b) Write the HTML code to design the web page as shown below considering the given specifications :

My Schedule

Name	Morning	Noon	Evening
Sahil	Running	Self Study	Gym
Abhay	Yoga	Tuition	
Neha	School		Dancing

- **Specification 1 :** The webpage code should follow the correct HTML structure. "**`My Schedule`**" should be displayed using the first level heading, and the background colour of the page should be light yellow.
- **Specification 2 :** Use the table header tag for the column headers.
- **Specification 3 :** The table should have exactly 4 rows and 4 columns, with the first row displaying the table header and the remaining rows for student data, as shown above.
- **Specification 4 :** The table should use rowspan and colspan, wherever required, as shown in the given table.

Ans

```
<body bgcolor="lightyellow">          <!--Specification 1a -->
 <h1>My Schedule</h1>                 <!--Specification 1b -->
 <table border="2">
   <tr>
     <th>Name</th>                    <!--Specification 2 -->
     <th>Morning</th>
     <th>Noon</th>
     <th>Evening</th>
   </tr>
   <tr>
     <td>Sahil</td>                  <!--Specification 3 -->
     <td>Running</td>
     <td>Self Study</td>
     <td rowspan="2">Gym</td>        <!--Specification 4a -->
   </tr>
   <tr>
     <td>Abhay</td>
     <td>Yoga</td>
     <td>Tuition</td>
   </tr>
   <tr>
     <td>Neha</td>
     <td colspan="2">School</td>     <!--Specification 4b -->
     <td>Dancing</td>
   </tr>
 </table>
</body>
```

(½ mark for correct `Specification 1a`)
(½ mark for correct `Specification 1b`)
(½ mark for correctly writing `table` tag)
(½ mark for correctly writing any 2 `<tr>` tags)
(½ mark for correctly writing any 2 `<th>` tags)
(½ mark for correctly writing any 2 `<td>` tags)
(½ mark for correctly using `rowspan`)
(½ mark for correctly using `colspan`)

	SECTION E **Case-Study** 2×4=8	
25.	Dr. Chitra is a Computer Science Professor at a University and was working on a research paper since last few months. She recently submitted her paper to a publication house. A few months after publishing her paper, she found that another researcher has published a similar paper in a different journal, using some portions of her work without giving her credit. Dr. Chitra is upset and wants to take legal action. Based on the above information, answer the following questions : (i) Are netiquettes being followed in the given scenario ? Justify your answer. (ii) The act of using someone's work as your own is known as _____. (iii) Which type of intellectual property law was violated by the other researcher by publishing Dr. Chitra's work without consent ? (iv) Suggest any one way to prevent intellectual property law violation in the context of academic research.	4
Ans	(i) No, netiquettes are not being followed in the given scenario because (Any one of the following), ● using someone else's work without giving credit is plagiarism. ● Netiquette requires honesty, respect and proper acknowledgement of other's work. ● Copying ideas or content without citation is unethical ● Copying ideas or content without citation is unfair ● Copying ideas or content violates academic integrity/professional standards. OR Any other valid justification	
	(½ mark for the answer) ***(½ mark for justification)***	
	(ii) Plagiarism **OR** IPR Violation	
	(1 mark for correct answer)	
	(iii) Copyright law **OR** IPR - Intellectual Property Rights	
	(1 mark for correct answer)	
	(iv) take permission before using copyrighted material **OR** cite sources properly **OR** use plagiarism detection tools **OR** There should be strong laws & punishments in place to prevent violation **OR** There should be several awareness programmes for masses to prevent violation **OR**	

	Any other correct way	
	(1 mark for writing each correct answer)	
26.	Irfan lives in a rural area with limited access to the Internet. His children struggle to attend online classes, as the area has poor Internet connectivity. On the other hand, in the nearby city, people have access to high speed Internet and are able to utilize services effectively. Irfan feels that his family is at a disadvantage due to the lack of access to digital infrastructure. Based on the above information, answer the following questions : (a) Unequal access to digital technology is called _____. (b) Mention any one reason for this unequal access to technology. (c) Mention any one initiative the government can take to reduce this unequal access. (d) How does limited access to the Internet affect children in terms of education and opportunities ?	4
Ans	(a) Digital Divide **OR** equivalent conveying the same meaning	
	(1 mark for correct answer)	
	(b) Lack of infrastructure in rural areas **OR** High cost of technology **OR** Low digital literacy **OR** Poverty **OR** Educational inequalities **OR** Language barrier **OR** Any other valid reason	
	(1 mark for any correct reason)	
	(c) Make the Internet alongwith other resources available **OR** Provide affordable/free internet access in rural areas **OR** Promote digital literacy training **OR** Any other valid initiative	
	(1 mark for mentioning any one appropriate initiative)	
	(d) Limited internet access restricts children to access online educational resources, reducing their learning opportunities. **OR** Any other correct answer	
	(1 mark for any correct answer)	